



Haptic and auditory cues: a study on independent navigation for visually impaired individuals

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Abstract

The research study aimed to evaluate the effectiveness of a personal device designed to assist blind users in navigating their environment. The wearable device consists of a stereo vision camera with structure illumination, a central control unit, a haptic belt, speakers, and a custom-made keyboard. The device uses haptic stimuli and auditory cues to provide environmental information and facilitate navigation. An important feature of the system is that the user can select operating modes and their settings while navigating. The procedures for processing the depth map, its segmentation and a mapping scheme for converting the segmented image into a matrix of haptic actuators are explained. The usability of the system was verified in static and mobility trials with 10 blind people. A post-trial survey provides valuable feedback from users, highlighting strengths of the system, such as its functionality and safety features, as well as areas for improvement, such as the intuitiveness of the user interface. The results of the study show that the combination of haptic and auditory modalities to present the environment can be an effective approach to building usable navigation aids for the visually impaired.

Keywords Blindness · Navigation · Accessibility · Assistive Technology · Haptic Feedback · Electronics Travel Aid

1 Introduction

It is estimated that by 2030, more than 2 billion people will be living with some form of disability, including visual impairment. Currently, many visually impaired people rely on human assistance to function in their daily lives, which limits their independence and places a burden on others. There has been much research into the development of Electronic Travel Aids (ETAs) to support independent mobility for visually impaired people [1]. ETAs have the potential to provide a technological alternative for visually impaired people to nav-

igate their environment independently [2]. However, existing ETAs have their limitations. Substituting the sense of sight with auditory or tactile interfaces, despite dynamic technological advances, is a challenging task that still awaits a breakthrough solution that would be widely accepted by the visually impaired community.

Firstly, there is a large mismatch between the information capacity that can be acquired and processed by vision compared to other human senses. Secondly, methods of converting visual information into auditory or haptic signals, also known as auditory and haptic displays, need to be carefully designed and require extensive training to teach the visually impaired to correctly interpret such non-visual techniques of presenting the environment. Our long experience in building ETAs leads us to believe that solutions should be designed to provide as much interaction as possible in the way the blind user can control the ETA [3, 4]. Such an interaction should provide two aspects of ETA functionality. Firstly, the blind user should be able to control the distance and/or angular range of the environment that would be presented non-visually, and secondly, the blind user should be able to decide on the amount of tactile or auditory information required in a particular navigational situation, e.g. when walking in a park these might be only a coarse represen-

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tation of the environment, whereas in a city environment a more detailed spatial information of the urban environment should be presented. Finally, it should be emphasized that any ETA designed for the blind should be developed in close cooperation with the target users. Only their critical and constructive comments can provide the designers of the system with invaluable guidance on how to solve the key problems of mobility for blind people [5].

Given these challenges, multimodal interaction emerges as a key strategy for effective ETA design. By combining complementary sensory channels, such as haptics and audition, systems can provide richer and more intuitive feedback. The concept of representing information in a modality-independent way, enables seamless integration of different types of cues, while careful design ensures that the user is not overloaded cognitively. Multimodal interfaces, therefore, offer a promising approach to delivering both continuous spatial information and context-sensitive messages, enhancing usability, safety, and independence for visually impaired users.

In this paper, we propose a novel approach to the design of ETAs by paying special attention to the user's interaction with the assistive device. We use both haptic and auditory modalities in a simultaneous way, taking care not to overload the user with one of these senses. We see this feature of the system as a much-needed approach to building ETAs that overcomes the limitations of other existing systems[6]. Our goal is to develop an ETA interface that provides users with a more intuitive and interactive way to explore their environment, while also enhancing their independence and confidence in mobility.

2 Related works

There have been many engineering efforts aimed at building assistive devices for the visually impaired. Comprehensive reviews of research into technological aids for the visually impaired are given in [1, 7–10]. In particular, significant efforts have been made to develop electronic travel aids (ETA), i.e. devices or systems to aid mobility and navigation of the visually impaired [11]. The range of reported approaches to building ETAs devices ranges from simple obstacle detectors to environmental imagers[6, 12]. However, a common feature of these systems is that they operate on the concept of sensory substitution. In fact, such systems can also be classified as augmented reality systems, in which new sensory signals (haptic or auditory) are created to enrich the information about the environment.

Sensory substitution is a use of healthy sensory modality to substitute the lost sense [5, 7]. This approach has been used in various ways, including sonifying text-to-speech technologies [13], sonifying graphics [3] and using haptic or tactile

displays to present the geometric layout of the environment [14]. The notable example is optophone or Kurtzweil's "reading machine" [15].

There are also other techniques for converting visual images into haptic or auditory modalities. One such method is audio-description, in which a verbal narrative comments on the visual information [16]. However, this requires human participation in interpreting the scene content. Other solutions are based on automatic converters of visual data into auditory signals [11] or haptic stimulations [17, 18].

In recent years, solutions using vibrations to convey non-visual information about the environment to the user have been particularly developed, especially when the auditory channel is already occupied. Haptic devices, such as vests, belts, and integrated handles, use vibrations to communicate spatial and directional information [6, 10]. The use of vibrational feedback goes beyond simple warnings. Researchers have developed an advanced taxonomy in which various vibration features, such as frequency, rhythm, and length, are used to convey specific semantic information [19, 20]. This allows a single device to communicate not only the presence of an obstacle but also its nature – for example, a change in floor texture or the direction of stairs. This advanced use of haptic feedback has been implemented in a variety of systems for navigation and guidance. For instance, the system described in [21] uses four motors in a helmet to vibrationally transmit navigation instructions, which are sent by a remote operator using a joystick. In a different approach, the Glide system conveys haptic information through the handle of a robot that guides the user [22]. It should be noted, that while information conveyed through vibration mechanisms is generally effective, prolonged use can lead to fatigue, which can impair the user's ability to properly interpret the transmitted data.

It is worth noting that also AI can assist in interpreting images [23]. For example, AI can identify objects on an image or determine their location within the scene. However, AI is not yet capable of participating in real-time warning and navigation, which requires high-speed processing and sensory data from sources such as video cameras or haptic sensors.

Unfortunately, few of the technical solutions to assist the visually impaired use interactive approaches to present non-visual information about the environment [24]. By interaction, we mean here the functionality of the device or a computer application that would allow the blind user to take control of the incoming stream of non-visual information presenting the environment. We postulate that this is one of the main reasons why many ambitious research and engineering efforts to build assistive devices have failed to gain practical acceptance among the visually impaired community. Few interactive assistive approaches for the visually impaired have been reported in the literature, e.g. in [25],

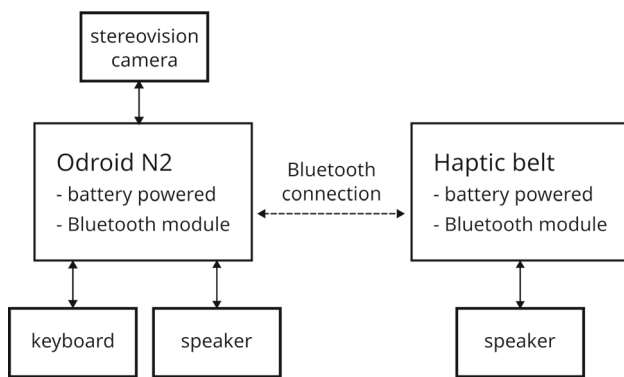


Fig. 1 Block diagram of the navigation assistance system for the visually impaired. Both devices are worn by a visually impaired person and are shown in Fig. 2

line elements of images were defined by the user for sonification, and in [26], a multi-level, user-controlled sonification approach was proposed.

3 System description

Unlike most of the assistive technologies being developed, our system focuses on incorporating interactive elements that allow users to customise their experience. For example, users will be able to switch between different modes of operation (the way obstacles are presented and the range of their presentation) or initiate device activation at will, tailoring the system to their specific needs and preferences. This level of personalisation is essential to ensure that people with visual impairments can effectively navigate and interact with their environment. In addition, we aim to design our system so that it fits seamlessly into everyday life, rather than drawing attention to the user's visual impairment. Unlike existing solutions, which often have conspicuous wearable components such as head-mounted displays or earpieces, our system can be hidden under clothing, with only the camera remaining visible. This minimizes visual cues and allows users to move freely without drawing attention to themselves.

The system consists of two main components: a central control unit built upon the Odroid-N2 single-board computer, which performs all necessary computations, ensuring the system's compact size and portability, and a haptic feedback device in the form of a wearable belt (Fig. 1).

The device is battery powered and has a 14-button tactile keyboard. Each key is equipped with a concave, tactile symbol. The keyboard enables the activation and deactivation of the device, selection of the operating mode, and allows for muting sounds and vibrations. Additionally, it provides the capability to put the device into a system parameter detection mode, where parameters such as the distance from the ground and the orientation of the stereoscopic camera rela-



Fig. 2 A photo of the system mounted on a visually impaired volunteer taking part in outdoor trials

tive to the ground are assessed. A stereo vision camera with structural illumination, the Intel RealSense D435i, is used for image capture. The combination of stereo vision and structural illumination enables the device to operate effectively in both bright daylight and at night. Tests have shown that the effectiveness of reconstructing a three-dimensional scene using structural lighting is significantly higher than using stereo vision alone, except in sunlight conditions, where structural lighting is less effective due to interference from ambient light [27]. The device runs on the Linux operating system and is also equipped with a loudspeaker for generating sound signals and voice messages. The loudspeaker integrated into the central control unit provides information about the selected operating mode and the status of the wireless connection with the haptic belt. A Bluetooth connection is used to exchange data with the haptic belt. The haptic belt is equipped with 22 vibrating elements (20 elements are used for obstacle localization presentation, while the remaining 2 provide information about special events, such as the detection of selected obstacle classes.) and, like the central device, is powered by rechargeable batteries. The inclusion of a speaker in the belt allows for checking the battery level (Fig. 2). The belt features a Bluetooth command decoder and actuator driver, based on an 8-bit microcontroller. The system features two speakers, one integrated into the central control unit and another within the wearable belt. This dual-speaker setup is a deliberate design choice, allowing each device to independently provide audio feedback, such as announcing its individual battery level. This functionality is crucial for the user, as it enables them to check the status of either device, even before the full system is assembled and connected.

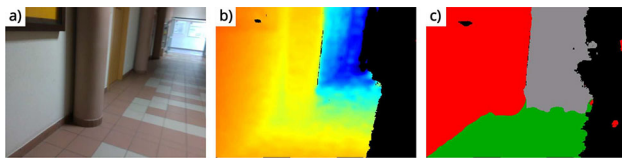


Fig. 3 Processing steps of the images. **a)** a single image of the corridor scene showing walls with columns and tiled floor, **b)** depth image of the corridor scene, with pseudocolours representing distance: the warmer the colour, the closer the object, **c)** corridor scene image segmented into regions recognised as floor (green), wall (red), the area outside the presentation range (gray), points for which the depth map cannot be determined (black)

4 Haptic presentation of the environment and device operation modes

The ground plane and obstacle detection algorithm (Appendix A) was implemented on Odroid N2, a single board computer. The algorithm requires a one-time interaction from the user; pressing a dedicated button initialises the automatic detection and tracking process by determining the device's distance to the ground and the camera's angle relative to it. The ground plane is recognized based on selected points in the scene defined in the depth maps as was described in Appendix A. Using the mathematical representation of the ground plane, it becomes possible to transform point coordinates into a ground-based coordinate system.

Subsequent algorithmic steps involve the identification of obstacles, i.e. objects that do not satisfy the ground plane equation (see Fig. 3). In addition, a method is proposed to track the ground plane in cases where it is not visible in the scene or is obscured by obstacle. We assume that the distance between the camera and the ground changes only slightly, as does the orientation of the camera relative to the ground. In this case, the equation of the ground from the previous frame is used for calculations, in addition to tracking the orientation of the camera relative to the ground using data from the stereo camera's built-in accelerometers.

We inform the user of any obstacle encountered along the travelled path, whether the obstacle touches the ground or not, such as a protruding branch, a pole or a step. By converting only the points identified as obstacles in the previous stage, it is possible to construct an occupancy grid (see Fig. 4) that shows the locations of obstacles projected onto the ground plane. The user is notified of obstacles within a pre-determined width range (0.8m, 1.2m or 1.5m). The smallest width corresponds to the user's body width, while the other two modes allow for a wider scanning area in front of the user. These broader scanning modes are useful for detecting open doors, narrow passages, and similar spatial features. This width range is divided into five sections (columns in the haptic actuator strip), each containing eight regions (twice the number of rows in the actuator strip). The number of

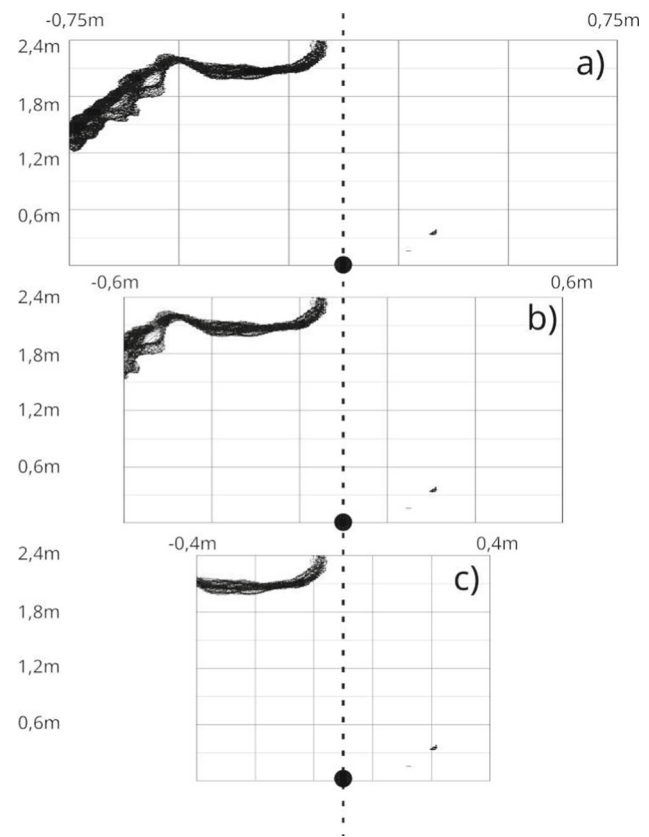


Fig. 4 Obstacle maps generated for scenes shown in Fig. 3 - with haptic image widths **a)** 1.5m, **b)** 1.2m, **c)** 0.8m. The scanning width is a configurable parameter that allows the user to adjust the detection range from a narrow width corresponding to the user's body to a wider range for identifying larger objects such as open doors or narrow passages. Visible elements of obstacles (objects whose points are farther than 10cm from the ground plane) are marked in black. The visually impaired person is positioned at the location indicated by the black dot. The visible rectangular grid on the screen divides the obstacle map into areas assigned to individual haptic actuators

pixels belonging to each region is counted to remove inaccuracies from the depth map values. The region containing obstacles closest to the user is then determined.

We previously conducted research to determine the optimal number of rows and columns for the vibrotactile actuator array. This involved testing the distinguishability of vibrotactile patterns across different matrix sizes, including 4×4 , 3×6 , 4×5 , 3×7 , and 4×6 . Based on these tests, we selected the matrix size that showed the best recognition of test patterns [28].

The device can operate in two modes, which differ in the way information is presented:

- basic mode - the device presents information about obstacles,

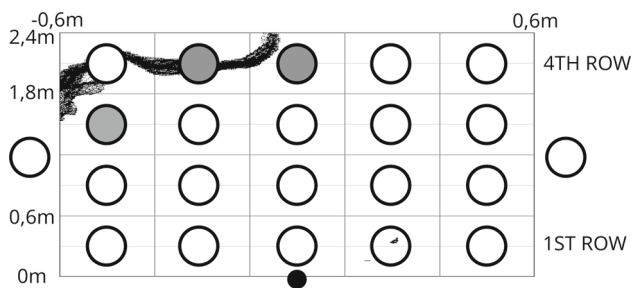


Fig. 5 The haptic presentation of the obstacle map for a width of 1.2m (view in the direction of the blind person's movement). Note that the belt is flipped vertically; in reality, closer elements are represented by actuators located higher in the band. For the visually impaired, we present only the closest elements in each column, making it easier to interpret the information and reducing energy consumption. The fourth actuator in the first row is not activated because the number of pixels corresponding to obstacles did not exceed the threshold. Three vibrating elements were activated, presenting information on the location of obstacles closest to the navigated person in each column

- free space mode - it presents information about the available space, which can be seen as the opposite or negation of the basic mode.

In basic mode, an actuator corresponding to this region is activated at a frequency determined by the location of the detected obstacle in the scene. In each column, only one actuator is activated when obstacles are present, reducing power consumption and making detection easier for the user. The closer the obstacle, the stronger the activation and the higher the frequency applied to the actuator; for example, standing approximately 2.5 metres from a wall triggers activation of the lower row of actuators, with successive rows activated as the user approaches the wall, and the top row activated when the user is approximately 50 cm from the wall. The vibration time varies according to the interval displayed: 100ms for closer areas and 50ms for further areas. Vibration frequency ranges from 30 to 65Hz, increasing in 5Hz increments. The haptic presentation of the obstacle map for a width of 1.2m is shown in Fig. 5 for the scene depicted in Figure 3.

The free space signalling mode is activated on request after selecting the corresponding button on the device. The system performs the same scene segmentation as for obstacle detection, then determines the sector where the obstacles are farthest from the navigated person if no obstacles are present. The stimulus associated with that distance and direction is then activated. In cases where obstacles are at the same distance or absent, the preferred direction indicating free space is the sector closest to the forward direction (i.e. the direction in which the blind person is moving). In other words, if objects are at the same distance or absent, the system presents information about the possibility of moving straight ahead (in the direction in which the blind person is moving) rather than suggesting an alternative path.

The detection of special objects such as stairs involved analyzing the orientation of the plane defined by points forming nodes of a grid overlaid on the lower part of the depth image. If a certain number of point triplets met the condition of orientation relative to the ground, it was assumed that visible steps were present in the scene. If stairs were detected in the scene, a dedicated actuator (not part of the matrix responsible for conveying obstacle information) was activated for the duration of the processing of one frame. If stairs were detected in every frame, a continuous signal is generated. An intermittent signal resulting from the detection of stairs not in every frame usually occurred due to a large distance from the stairs. In addition to haptic feedback, the presence of stairs and special objects of interest was also communicated through auditory messages, thereby strengthening the multimodal nature of the interaction.

All system settings, such as the mode of operation, were communicated to the user via auditory messages (speech synthesis) and short signals emitted by the device. This ensured that the user was left in no doubt as to the mode change that had occurred. Similarly, notifications were provided when the device started up and established a wireless connection with the belt, when the battery level was low, or when the vibrating elements activated the 'mute' mode. This mode is useful, for example, during a conversation with another person who, from the point of view of the image analysis algorithms, acts as an obstacle when standing opposite the system user.

In line with the principles of multimodal interaction, the system integrates both haptic and auditory modalities to facilitate navigation. The haptic belt delivers continuous spatial information, whereas auditory messages serve several functions: reporting the presence and attributes of special objects (such as the color of a traffic light or the direction of stairs), confirming user choices in the menu, and providing spoken distance measurements to the nearest obstacle upon request. This dual-channel approach ensures redundancy, improves intuitiveness, and increases user confidence in dynamic environments.

5 Testing procedure

The testing procedure was designed to balance controlled experimental conditions with realistic scenarios encountered by visually impaired users. The static phase enabled us to isolate and evaluate basic obstacle detection accuracy, while the mobility phase simulated real-life navigation challenges in both structured and unstructured environments. This two-step design was inspired by common practices in orientation and mobility training for blind individuals, where tasks progress from simple detection exercises to complex navigation tasks. Similar procedures are frequently adopted in



Fig. 6 Google Map showing the location and route of the tests. The visually impaired route is marked with a red line

studies evaluating assistive technologies for blind navigation [29, 30].

Recruitment of participants for the trials was carried out in cooperation with the Polish Association of the Blind. The tests were approved by the Bioethics Commission of the Medical University of Lodz, Poland, and took place in a designated area of the park on the campus of the Lodz University of Technology, where conditions similar to real-life use were simulated. In the initial phase of the study, the participants were familiarised with the operation and control of the device, in particular the control panel that allowed the manipulation of the device. All subjects received a detailed explanation of the test procedure and were asked to sign an informed consent form.

The test procedure consisted of two phases:

Static tests

The subject stood in a stationary position. Obstacles were placed at different distances from the participant. The participant's task was to indicate the obstacle with their hand and to determine its distance. Vibrations corresponding to the position of the obstacle were generated by belt actuators. The haptic belt was positioned on the participant's abdomen.

Mobility tests

TEST 1: The test setup consisted of a 100-metre route, divided into two distinct sections. The first 40 metres featured a straight path with three pre-placed obstacles in the participant's path: a person, a chair, a board (Fig. 6). Vibrations corresponding to the position of the obstacle were felt

on the abdomen. The subject's task was to identify the location and distance of an obstacle.

The second part of the route, approximately 60 metres, offered a more natural and varied terrain for a relaxed walk. Here, the subject encountered spontaneous obstacles typical of an outdoor environment, such as trees, a no-walk bollards, and stairs. This section aimed to simulate everyday encounters with unexpected obstacles while moving freely.

TEST 2: The subject walked along a 20 metre straight path in an open space and another person quickly entered the path. The participant stopped when they detected the fast moving obstacle.

TEST 3: Detection of stairs and surface irregularities. Participants were asked to identify stairs leading upward within their field of view.

TEST 4: Detection of free space. Clearance detection. The device had a free space detection function that worked in reverse to the obstacle detection function. Vibrations on the abdomen indicated the presence of free space between obstacles. The participant's task was to navigate in the direction indicated by the device.

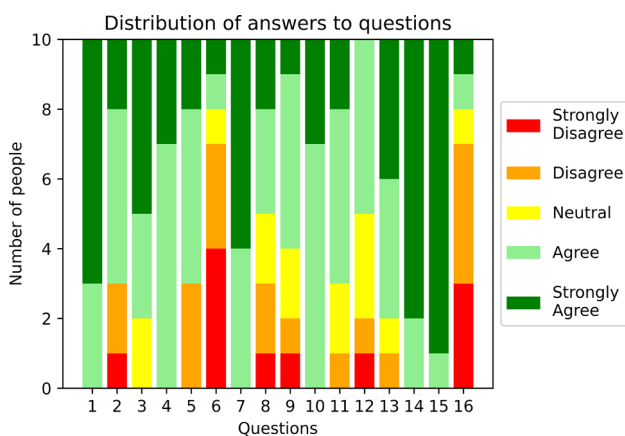
During the tests, information about detected obstacles was continuously provided to the subject by the device. During the mobility tests, in addition to obstacle detection and avoidance, participants were asked to report the time at which they perceived stimuli indicating proximity to an obstacle and the time at which they perceived an obstacle directly in front of them. Participants maintained constant voice contact with the experimenter. The main tests were preceded by a short training session. After completion of the tests, an interview was conducted to assess opinions on the usefulness of the tested prototype. During the interview, questions were asked based on the System Usability Scale (SUS) [31], focusing in particular on the detector set and the device's control panel. The research questions (see Table 1) were validated by an Orientation and Mobility (O&M) instructor from the Polish Association of the Blind.

6 Survey results

A total of 10 participants aged between 37 and 56 years (7 women and 3 men, average age: 44 years) took part in the study (both blind and severely visually impaired, all members of the Polish Association of the Blind). The quantitative analysis focused on the accuracy of reading information about the location of the obstacle and the ability to avoid the obstacle without physical contact. The reading of obstacle location information was flawless in all study participants. Dynamic tests demonstrated 100% effectiveness of the tested device under simulated conditions. It is worth noting that the system was assumed to detect obstacles of a minimum size of approximately 20 cm, without considering the detection of

Table 1 System Evaluation Survey Questions

Q1	Is it necessary to have voice messages (belt and chest module) to inform about the battery level?
Q2	Are the sound messages emitted by the belt audible?
Q3	Is the voice of the messages pleasant?
Q4	Is the belt comfortable to wear?
Q5	Is the chest module comfortable to wear?
Q6	Are the buttons easy to feel and recognize?
Q7	Would convex buttons be better?
Q8	Are the stimuli emitted by the belt regarding the detection of curbs or steps distinguishable from stimuli regarding the detection of other obstacles?
Q9	Are the stimuli emitted by the belt to detect kerbs or steps legible and appropriately selected?
Q10	Was obstacle avoidance easy?
Q11	Are the stimuli emitted by the belt, warning of obstacles and indicating the location of obstacles in space, readable and appropriately chosen?
Q12	Is the assignment of functions to buttons intuitive and logical?
Q13	Should vibration stimuli be supported by sounds or verbal information?
Q14	Is the mute button (interrupting stimuli) useful?
Q15	Is the kerb or step detection function important and necessary?
Q16	Is the risk associated with additional stimuli for presenting information about special object categories minimal, and their interpretation unambiguous, and not misleading?

**Fig. 7** A graph showing individuals' responses to survey questions from Table 1. Responses are colour coded on a scale from strongly disagree to strongly agree

uneven surfaces or gaps in the pavement. The qualitative analysis consisted of surveys completed by the study participants, with the questions listed in Table 1. The results are presented in the graph shown in Fig. 7.

7 Discussion

The results of the survey conducted with the 10 trial participants shed light on various aspects of the usability and effectiveness of the device under investigation. Overall, the

results indicate a generally positive reception of the device's functionalities and safety features, with room for refinement of the user experience and design elements. One notable aspect is the perceived need for voice messages to inform users of the battery charge status, with the majority of participants rating this feature positively. Similarly, the sound messages emitted by the belt are generally considered to be audible, although opinions are more divided on the pleasantness of the voice of these messages. Comfort of use is another important consideration, with most participants finding both the belt and the chest module comfortable to wear. However, opinions on the tactile and recognisability aspects of the buttons are mixed, suggesting potential areas for improvement in button design. The majority of participants prefer large, convex buttons, suggesting possible ways to improve user interaction.

The survey also reveals positive perceptions of the clarity and appropriateness of the stimuli emitted by the belt for detecting kerbs or steps, and for warning of obstacles and indicating their position in space. However, there are suggestions that the vibration stimuli could be supported by sound or verbal information, suggesting potential improvements in providing feedback to users. The mute button, which allows the interruption of stimuli, was found useful by most participants, highlighting its importance for comfort in everyday use. In addition, the kerb or step detection feature was considered important and necessary by the majority of participants. Contrary to the initial assumption, the participants generally did not perceive the risk associated with additional stimuli for

the detection of special object categories as minimal. Despite the use of two additional actuators, the way these categories were presented could still be confused with regular obstacles, and there were suggestions that the presentation should be made even more distinguishable. Nevertheless, these observations reflect positively on the safety aspects of the device.

Although the survey highlights several strengths of the device, such as its functionality and safety features, it also identifies areas for improvement, particularly in button design and user interface intuitiveness. These findings provide valuable insights for refining and improving the design of the device, ultimately enhancing the user experience and usability.

8 Summary

This article focuses on the evaluation of a method for presenting environmental information to blind users using a device equipped with haptic stimuli, supplemented with auditory cues for exploring selected device operating modes. The system integrates a stereo vision camera enhanced by an IR projector (Intel®RealSense™D435) for depth perception. It maps the environment onto an occupancy grid, dividing the depth of the scene into cells to determine the presence of objects. Haptic feedback is provided by a custom-designed tactile belt equipped with 20 vibrating actuators arranged in a 4x5 grid, and 2 additional actuators to provide additional information about the presence of special categories of objects. Through surveys and tests, the research aimed to assess blind users' acceptance of this approach to independent navigation. Overall, the results indicate a generally positive reception of the functionalities and safety features of the device, with potential for improvement in user interaction and design aspects. The research highlights the potential of haptic and auditory cues to enhance navigation for the visually impaired, while also highlighting areas for improvement in device design and user interaction.

By incorporating interactive elements and designing the system for seamless integration, we hope to create a more inclusive environment where people with visual impairments can participate on an equal footing with their sighted peers. Our goal is to empower the visually impaired users to take control of their surroundings, make informed decisions, and engage fully in social interactions, all while maintaining their independence and dignity.

Appendix A. Depth map processing and scene segmentation

In mobility assistance systems for the blind, the detection and tracking of the ground plane is crucial as it serves as

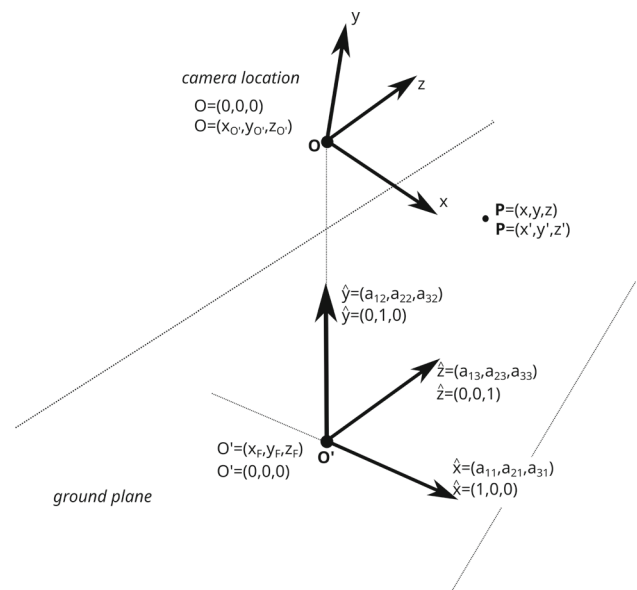


Fig. 8 Geometric arrangement of the camera coordinate system (O) and the ground-aligned coordinate system (O'). The coordinate systems O and O' , point coordinates P , and basis vectors are represented in the camera coordinate system (top line) and the ground-related coordinate system (bottom line), respectively

a reference for identifying obstacles in the travelled path. Detailed procedures for processing image data related to ground detection and tracking have been described in [32] and [33].

Our experiments indicated that the camera's pitch angle should be slightly less than half of its vertical field of view. This ensures visibility of both the ground plane and objects above the user's head. Given this setup, obstacle distances are best defined in a coordinate system aligned with the ground plane (O') rather than the camera's original coordinate system (O), as shown in Figure 8.

Let the equation of the ground plane be $Ax + By + Cz + D = 0$, and let the point O' be the orthogonal projection of the origin of the camera coordinate system O onto this plane. The new origin of the O' coordinate system, expressed in the O coordinate system, has coordinates $[x_F, y_F, z_F]$, which are determined by assuming that the origin of the camera coordinate system is the closest point to the ground plane. The unit basis vectors of the O' coordinate system in the camera coordinate system are illustrated in Fig. 8. The basis vector $\hat{y}$ is aligned with the normal vector to the ground plane $[A, B, C]$. The basis vector $\hat{z}$ follow the direction in which scene depth is measured. Finally, $\hat{x}$ is obtained using the cross product: $\hat{x} = \hat{z} \times \hat{y}$.

The coordinates of any point P can be transformed between the coordinate systems O and O' using equations (A1) and (A2).

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x_F \\ y_F \\ z_F \end{bmatrix} + \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} \quad (\text{A1})$$

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} x_{O'} \\ y_{O'} \\ z_{O'} \end{bmatrix} + \begin{bmatrix} a_{11} & a_{21} & a_{31} \\ a_{12} & a_{22} & a_{32} \\ a_{13} & a_{23} & a_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad (\text{A2})$$

By applying Eq. (A2), the coordinates of each point in the scene can be determined within the O' coordinate system, which is aligned with the ground plane. Based on these coordinates, an obstacle map, known as an occupancy grid, can be constructed.

The ground plane estimation algorithm begins with an initialization phase, during which it determines the camera's distance from the ground as well as its tilt relative to the ground plane. This initial setup requires a clear, unobstructed view of the ground to ensure accurate calibration. The ground plane is estimated using the farthest points in the lower part of the depth map, assuming the camera is level with the ground and its roll angle is close to zero. These initial parameters provide a robust initial approximation of the ground plane.

Once the camera's orientation and distance are determined, the system transitions to the ground plane determination phase. The depth image is divided into triangular sections using a regular grid. The equation of the plane for each section is calculated based on the coordinates of its vertices. A plane is considered part of the ground if its distance from the camera differs by no more than 30 cm from the initial estimate and if the angle between its normal vector and the initial ground normal is within 25 degrees. The selected points are collected to refine the ground plane estimation.

If significant deviations occur due to obstacles or large objects like walls exceeding 20 cm in distance or 20 degrees in orientation the system retains the previously saved plane for further calculations. This ensures stability in navigation even in environments with temporary or partial occlusions.

The effectiveness of the proposed method for ground plane detection and tracking was verified by placing a camera on a tripod in an indoor scene. For the visible ground plane, the ground truth plane equation was determined without obstacles, and the scene was segmented using a depth map. This map was then treated as ground truth. Next, obstacles were placed in the scene, and the ground plane was determined for the current frame S_i . The Jaccard index was used as a comparative measure.

For each frame of the scene, two segmentation maps were generated: one for the ground truth plane SGT_i and one for the currently detected plane S_i . The ground truth plane was determined using the equation obtained from the obstacle-free scene, while segmentation was performed for the analyzed frame with obstacles.

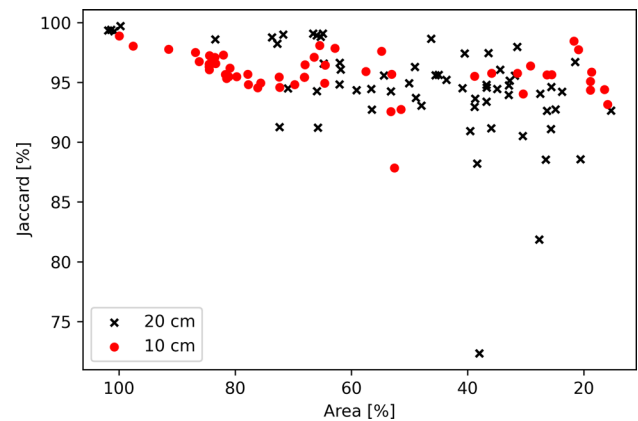


Fig. 9 Verification of the Ground Plane Estimation Method. Dependence of the Jaccard index on the visible ground surface for two acceptance thresholds (10 cm and 20 cm), defined as the maximum distance of points from the ground plane at which they are still considered part of the ground

$$J(SGT_i, S_i) = \frac{|SGT_i \cap S_i|}{|SGT_i \cup S_i|} \quad (\text{A3})$$

where:

- SGT_i is the set of pixels from the ground truth map for frame i ,
- S_i is the set of detected ground plane pixels for frame i ,

The area [%] metric represents the proportion of the detected ground plane in the scene with obstacles relative to the empty scene, based on the reference equation. To determine whether a point belongs to the ground plane, a distance threshold was applied. A point was considered part of the plane if its distance from the reference plane equation was within 10 cm or 20 cm, depending on the chosen acceptance criterion (Fig. 9). The ground occlusion factor (Area) is defined as the ratio of the ground surface area in the obstacle-free scene to the ground surface area in the scene with obstacles, calculated based on the reference (ground truth) plane equation. This factor quantifies the extent to which obstacles obscure the ground, with higher values indicating a greater proportion of the ground being visible.

Figures 10 and 11 show, respectively, the dependence of the angle between normal vectors for the plane determined by the ground truth equation and the one estimated in the scene as a function of the ground occlusion factor, as well as the dependence on the camera-to-ground distance.

As can be observed from the charts (Figs. 9, 10, 11), the estimation of the ground plane equation is achievable even when it is significantly obscured by obstacles. However, there are situations where errors in estimation occur. But even in such cases, it is possible to use the equation correctly determined in previous time frames.

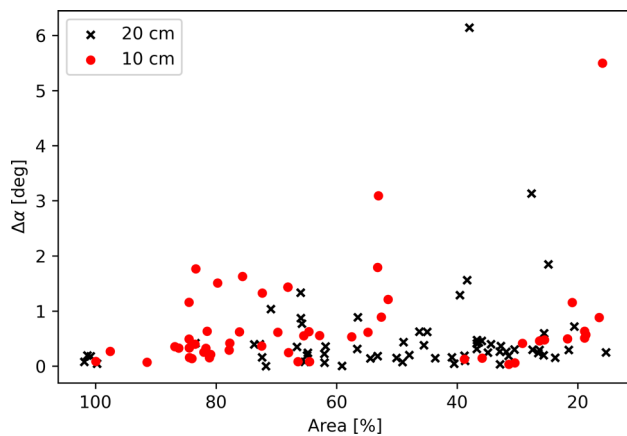


Fig. 10 Dependence of the angle between normal vectors for the ground truth plane and the detected plane on the ground occlusion factor. The results are presented for two acceptance thresholds: 10 cm and 20 cm

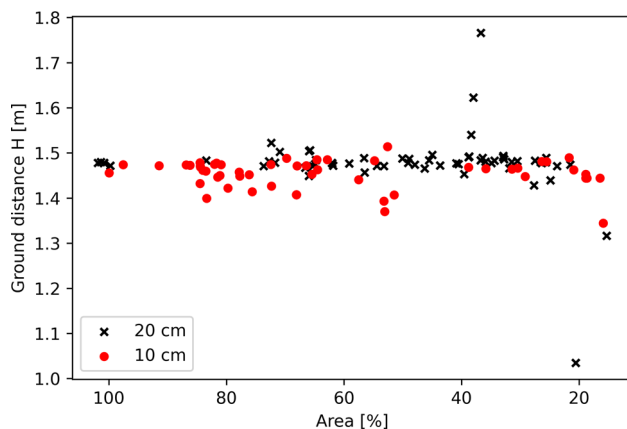


Fig. 11 Dependence of the camera-to-ground distance on the ground occlusion factor. The results are presented for two acceptance thresholds: 10 cm and 20 cm

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Declarations

Competing interests The authors declare no competing interests.

Ethics Approval and Consent to Participate The study involving visually impaired participants was approved by the Bioethics Commission of the Medical University of Lodz, Poland.

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